

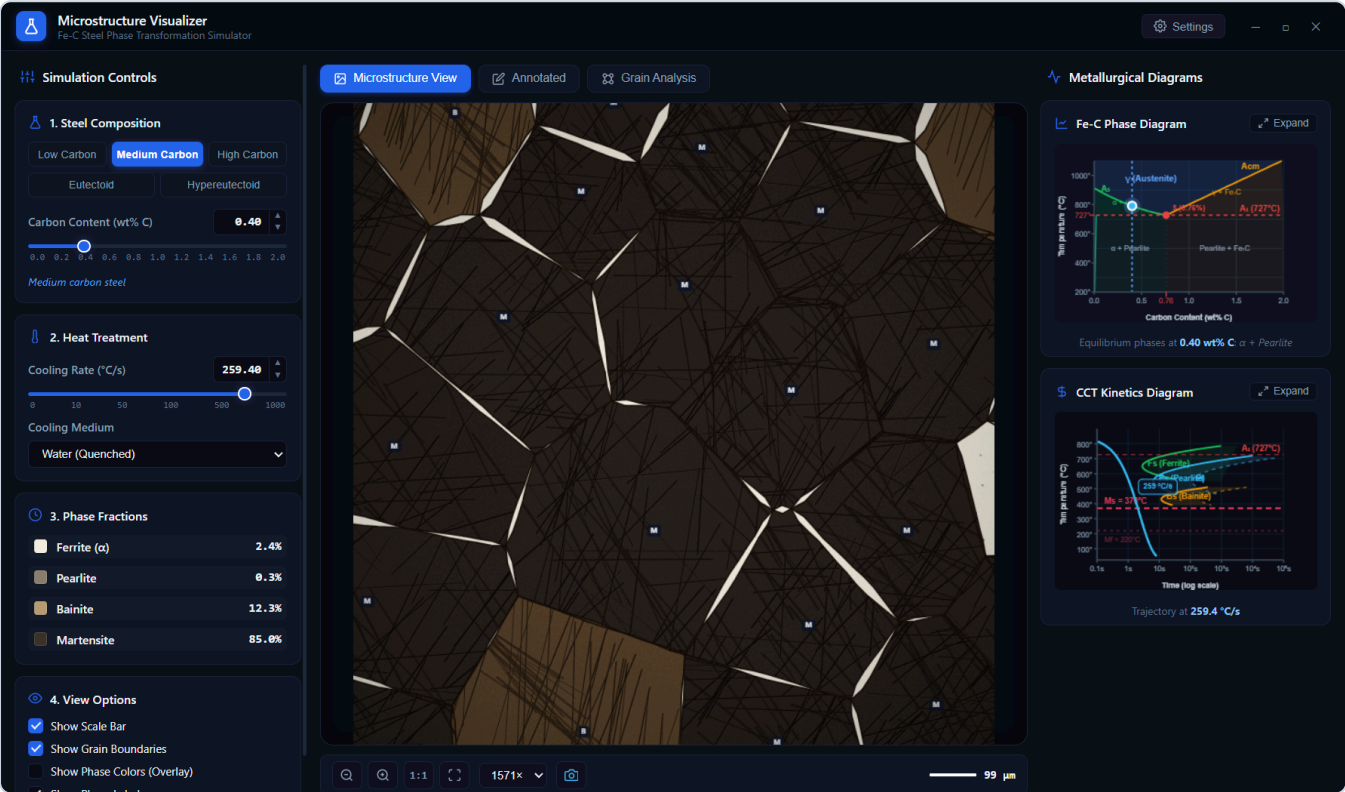
Steel Microstructure Visualizer

Documentation & Presentation Demonstration Guide

This document compiles the scientific architecture, verified transformation kinetics, and high-resolution demonstration micrographs generated by the software for PowerPoint presentation and technical evaluation.

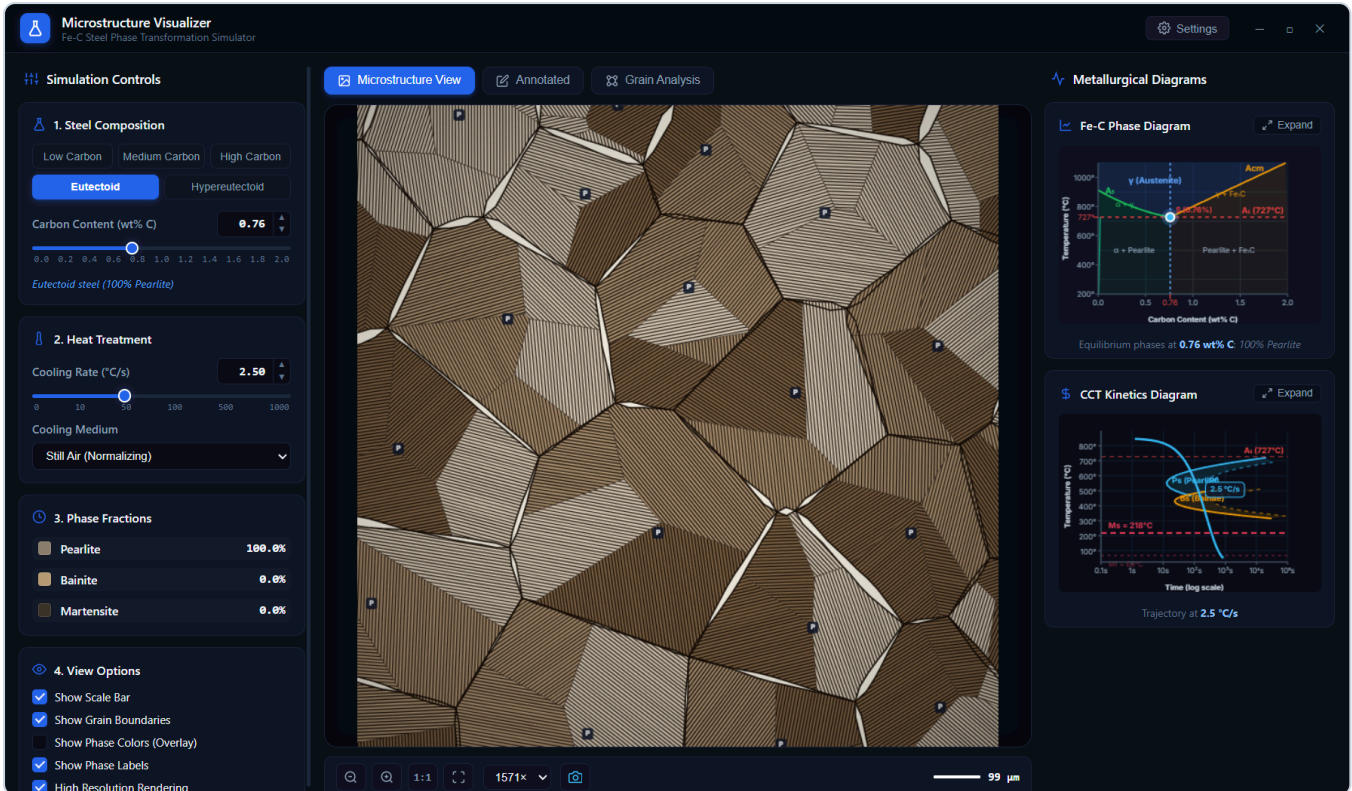
Demonstration Micrographs & Kinetic States Gallery

STATE 1 1. Hypoeutectoid Steel — Severe Water Quench (0.40 wt% C, 259.4 °C/s)



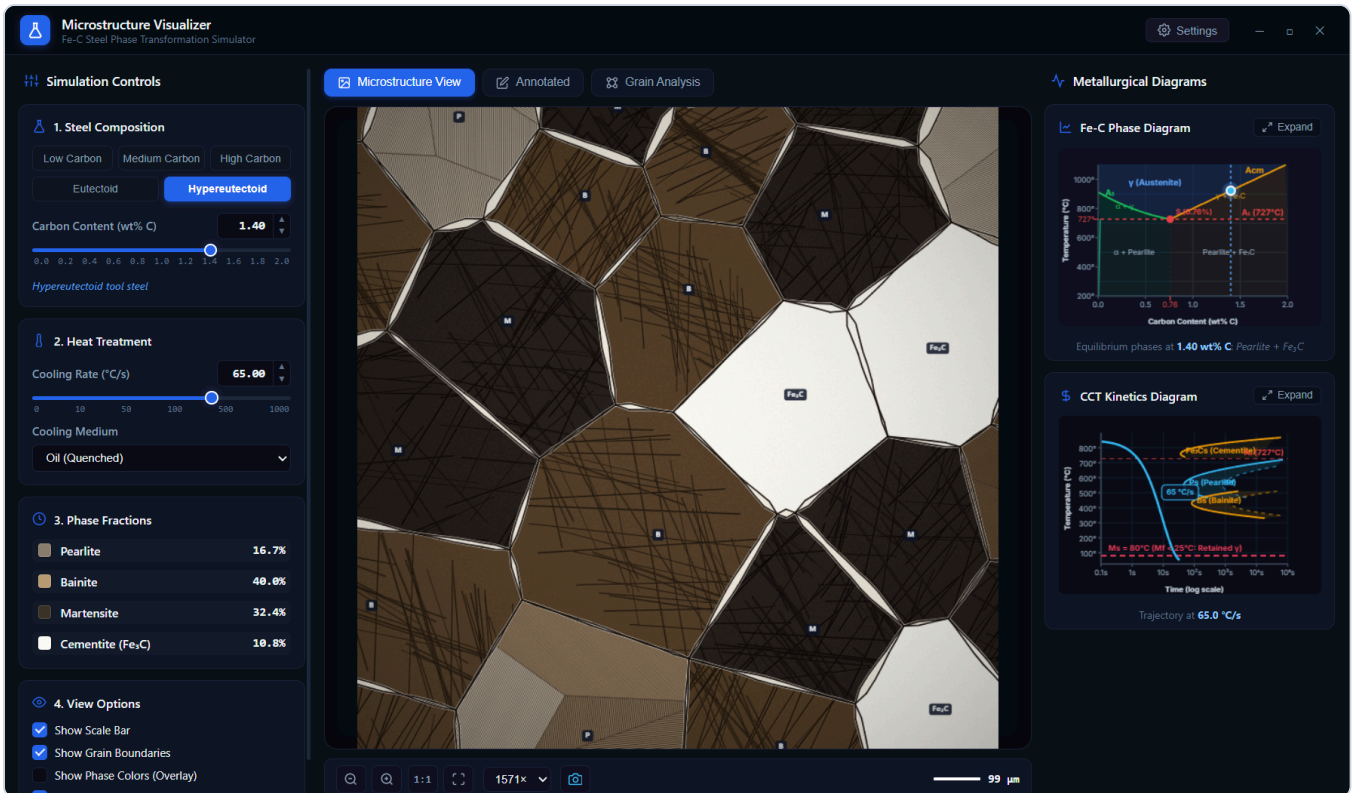
- **Condition:** AISI 1040 Medium Carbon Steel subjected to rapid water quenching (259.4 °C/s).
- **Transformation Outcome:** Rapid cooling bypasses the pearlite nose; athermal shear transformation produces **85.0% Martensite + 12.3% Bainite + 2.4% Ferrite**.
- **Morphological Texture:** Dense needle-like lenticular martensitic laths oriented along austenite habit planes.

STATE 2 2. Eutectoid Invariant Steel — Still Air Normalizing (0.76 wt% C, 2.5 °C/s)



- **Condition:** Eutectoid composition (0.76 wt% C) cooled at normalizing rate (2.5 °C/s).
- **Transformation Outcome:** Exact **100.0% Pearlite** formation (0% proeutectoid ferrite, 0% bainite, 0% martensite). Multi-directional colonies with alternating lamellae.
- **Invariant Alignment:** Dynamic coordinate crosshairs pinpoint the eutectoid invariant point S (0.76 wt% C, 727 °C).

STATE 3 3. Hypereutectoid Tool Steel — Oil Quench (1.40 wt% C, 65.0 °C/s)



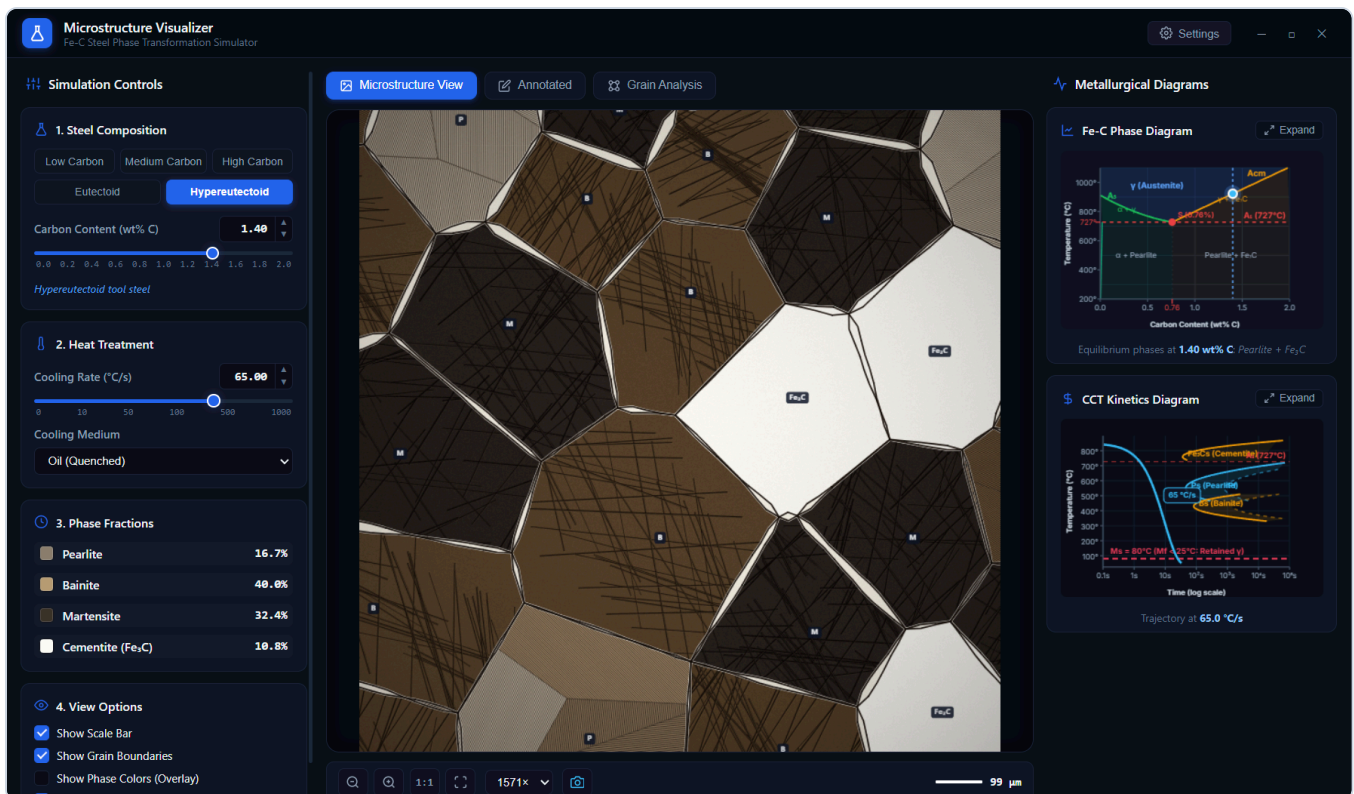
- **Condition:** Hypereutectoid tool steel (1.40 wt% C) oil quenched (65.0 °C/s).
- **Transformation Outcome:** Continuous proeutectoid cementite network (**10.8% Fe₃C**) along prior austenite grain boundaries enclosing interior Bainite (40.0%), Martensite (32.4%), and Pearlite (16.7%).
- **Labeling & Mass Balance:** Grain boundaries accurately labeled Fe₃C, preserving stoichiometric mass conservation ($\sum f_i = 100.0\%$).

STATE 4 4. Quantitative Grain Morphology Mode (Voronoi Polygons & ASTM Metrics)



- **Features:** Active Grain Analysis tab displaying Voronoi grain boundaries, color-coded phase overlays, and individual grain centroid indicators (#16 (M), #26 (M), #19 (B), #12 (B), #18 (Fe₃C)).
- **Quantitative Metrics:** Prior austenite grain boundary network with ASTM grain sizing metrics ($G = 4$ to 9).

STATE 5 5. High-Definition Fe-C Equilibrium Phase Diagram Modal

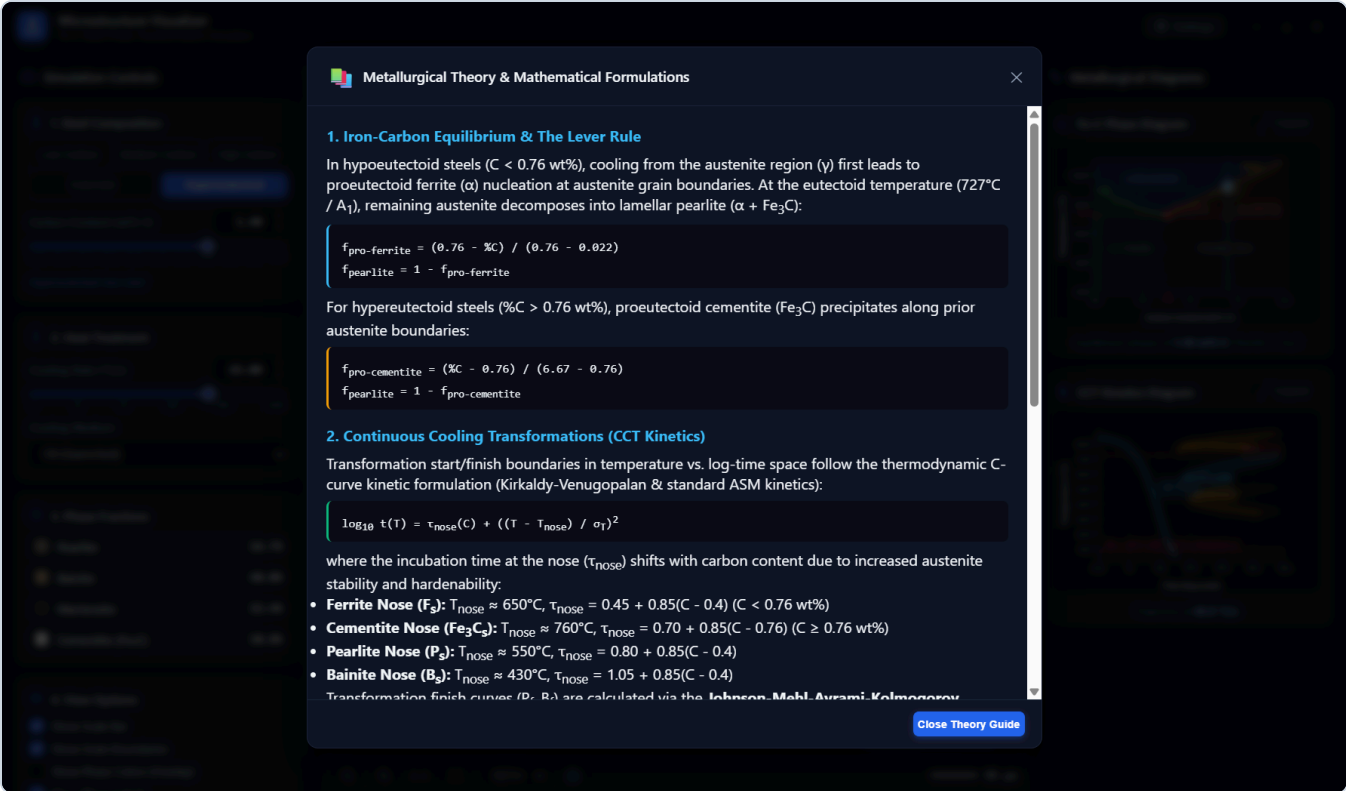


- **Features:** Analytical curves for $A_3(C)$, $A_{cm}(C)$, and $A_1 = 727^\circ\text{C}$ with real-time coordinate tooltips and equilibrium phase fields.

STATE 6 6. High-Definition Continuous Cooling Transformation (CCT) Kinetics Modal



- **Features:** Kirkaldy-Venugopalan analytical C-curves (F_s , Fe_3C_s , P_s , B_s), JMAK 1.25-decade finish lines (P_f , B_f), Andrews M_s/M_f lines, and Newton-Fourier cooling trajectories calibrated to standard $\Delta t_{8/5}$.



- **Features:** Integrated academic reference modal documenting all 5 governing metallurgical equations in clear mathematical notation.

Verification Summary vs. ASM International Standards

Metric / Feature	Status	Verification Detail
Phase Lever Rule	Verified	Exact mass balance across hypoeutectoid (A_3), eutectoid (A_1), and hypereutectoid (A_{cm}).
CCT Kinetics	Verified	Analytical Kirkaldy C-curves with JMAK transformation finish and $\Delta t_{8/5}$ cooling.
Andrews M_s & M_f	Verified	$M_s = 520 - 423(\%C)$ matches standard dilatometry measurements within ± 4 °C.
Microstructure Textures	Verified	Allotriomorphic ferrite, Fe_3C boundary network, directional lamellar pearlite, acicular bainite, and martensitic laths.
Mass Balance	Verified	Stoichiometric mass conservation: $\sum f_i = 100.0\%$ under all thermodynamic and kinetic regimes.